Introduction:

The real number system had limitations that were at first accepted and later overcome by a series of improvements in both concepts and mechanics. In connection with quadratic equations, we encountered the concept of imaginary number and the device invented for handling it, the notation

$i^2 = -1 \ or \ i = \sqrt{-1}$

In this lecture we continue the extension of the real number system to include imaginary' numbers. The extended system is called the complex number system.

Complex Number

A complex number is a number of the form $a + bi$, where a and b are real and $i^2 = -1 \ or \ i = \sqrt{-1}$. The letter '$a$' is called the real part and 'b' is called the imaginary part of $a + bi$. If $a = 0$, the number bi is said to be a purely imaginary number and if $b = 0$, the number a is real number. Hence, real numbers and pure imaginary numbers are special cases of complex numbers. The complex numbers are denoted by Z, i.e.,

$$Z = a + bi$$

In coordinate form $\quad Z = (a, b)$

<u>Note</u>: Every real number is a complex number with 0 as its imaginary part.

<u>Note:</u> Electrical engineers often write j instead of i because current is usually denoted by i.

Properties of Complex Number:

(i) The two complex numbers $a + bi \ and \ c + di$ are equal if and only if $a = c$ and $b = d$ for example if

$$x - 2 + 4yi = 3 + 12i$$

$$Then\ x-2=3 \quad and \quad y=3$$

(ii) If any complex number vanishes then its real and imaginary parts will separately vanish.

For example, if

$$a+bi=0\ then\ a=-bi$$

Squaring both sides

$$a^2=-b^2\ or\ a^2+b^2=0$$

Which is possible only when *a = 0, b = 0.*

Basic Algebraic Operation on Complex Numbers

There are four algebraic operations on complex numbers

(i) Addition:

$$\text{If } Z_1=a_1+b_1i \quad and\ Z_2=a_2+b_2i\ \ then$$

$$Z_1+\ Z_2\ =(a_1+b_1i)+(a_2+b_2i\ )=(a_1+a_2)+i(b_1+b_2)$$

(ii) Subtraction:

$$\text{If } Z_1=a_1+b_1i \quad and\ Z_2=a_2+b_2i\ \ then$$

$$Z_1-\ Z_2\ =(a_1+b_1i)-(a_2+b_2i\ )=(a_1-a_2)+i(b_1-b_2)$$

(iii) Multiplication:

$$\text{If } Z_1=a_1+b_1i \quad and\ Z_2=a_2+b_2i\ \ then$$

$$Z_1\,Z_2\ =(a_1+b_1i).(a_2+b_2i\ )=a_1a_2+a_1b_2i+b_1a_2i+(b_1b_2)i^2$$

$$Z_1\,Z_2\ =a_1a_2-b_1b_2+i\ (a_1b_2+b_1a_2)$$

(iv) Division:

$$\frac{Z_1}{Z_2}=\frac{a_1+b_1i}{a_2+b_2i}$$

Multiply Numerator and denominator by the number (a_2-b_2i) in order to make the denominator real.

$$\frac{Z_1}{Z_2} = \frac{a_1 + b_1 i}{a_2 + b_2 i} \times \frac{a_2 - b_2 i}{a_2 - b_2 i} = \frac{a_1 a_2 + b_1 b_2 + i\,(b_1 a_2 - a_1 b_2)}{{a_2}^2 + {b_2}^2 + i\,(a_2 b_2 - a_2 b_2)}$$

$$\frac{Z_1}{Z_2} = \frac{a_1 a_2 + b_1 b_2 + i\,(b_1 a_2 - a_1 b_2)}{{a_2}^2 + {b_2}^2} = \frac{a_1 a_2 + b_1 b_2}{{a_2}^2 + {b_2}^2} + i\,\frac{(b_1 a_2 - a_1 b_2)}{{a_2}^2 + {b_2}^2}$$

Generally, result will be expressed in the form *a* + *ib*.

Examples:

If $Z_1 = 8 + 3i \quad and\; Z_2 = 9 - 2i$

$Find\ (Z_1 + Z_2), (Z_1 - Z_2), (Z_1.Z_2)\ and\ (\frac{Z_1}{Z_2})$

Sol.

1. $Z_1 + Z_2 = 17 + i$

2. $Z_1 - Z_2 = -1 + 5i$

3. $Z_1.Z_2 = (72 + 6) + i(-16 + 27) = 78 + 11i$

4. $\frac{Z_1}{Z_2} = \frac{8+3i}{9-2i} = \frac{8+3i}{9-2i} \times \frac{9+2i}{9+2i} = \frac{72-6+27i+16i}{81+4} = \frac{66}{85} + \frac{43}{85}i$

Find the values of the following:

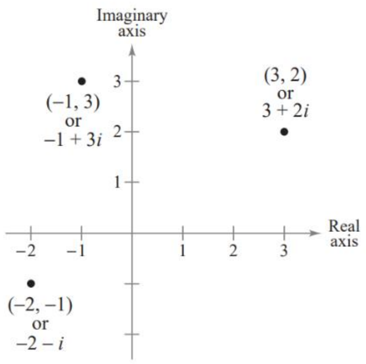
1. $\frac{1+5i}{-3i}$ 2. $\frac{7-i}{2+10i}$ 3. $2i + (-8 - 15i)$ 4. $(6 + 2i)^2$ 5. $4 - (3 - 20i)$

6. $(10 + 3i)(-1 + 7i)$

<u>Polar Form of a Complex Number</u>

Just as real numbers can be represented by points on the real number line, you can represent a complex number $Z = a + bi$

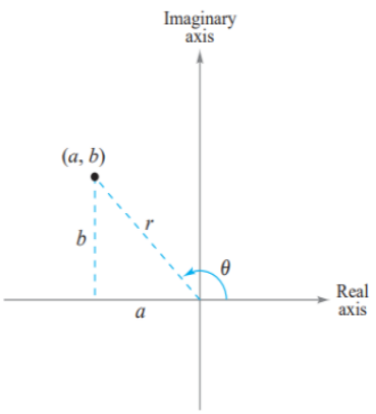
at the point (a,b) in a coordinate plane (the **complex** plane). The horizontal axis is called the **real axis** and the vertical axis is called the **imaginary axis**, as shown in figure below.



The **absolute value** of the complex number a+bi is defined as the distance between the origin (0,0) and the point (a,b) and is calculated as follows:

$$|a + bi| = \sqrt{a^2 + b^2}$$

To work effectively with *powers and roots* of complex numbers, it is helpful to write complex numbers in **polar form**. In the figure below, consider the nonzero complex number a+bi



By letting θ be the angle from the positive *x-axis* (measured counter-clockwise) to the line segment connecting the origin and the point (a,b) you can write

$$a = r\,cos\,\theta \ \ and\ b = r\,sin\,\theta$$

$$\text{where } r = \sqrt{a^2 + b^2}$$

from which you can obtain the **polar form** of a complex number.

Consequently, you have

$$Z = a + bi = (r\,cos\,\theta) + (r\,sin\,\theta)i = r(cos\,\theta + i\,sin\,\theta)$$

$$\tan\theta = \frac{b}{a}$$

The number r is the **modulus** of Z, and θ is called an **argument** of Z.

<u>NOTE</u> The polar form of a complex number is also called the **trigonometric form**. Because there are infinitely many choices for θ, the polar form of a complex number is not unique. Normally, θ is restricted to the interval

$$0 \leq \theta < 2\pi$$

although on occasion it is convenient to use $\theta < 0$.

Example:

Write the following complex number in polar form

$$Z = -2 - 2\sqrt{3}\,i$$

Sol.

$$Since\ Z = a + bi = -2 - 2\sqrt{3}\,i\ then\ a = -2\ and\ b = -2\sqrt{3}$$

The absolute value of Z is

$$r = |a + bi| = \sqrt{a^2 + b^2} = \sqrt{4 + 12} = 4$$

And the angle θ is given by

$$\tan\theta = \frac{b}{a} = \frac{-2\sqrt{3}}{-2} = \sqrt{3}$$

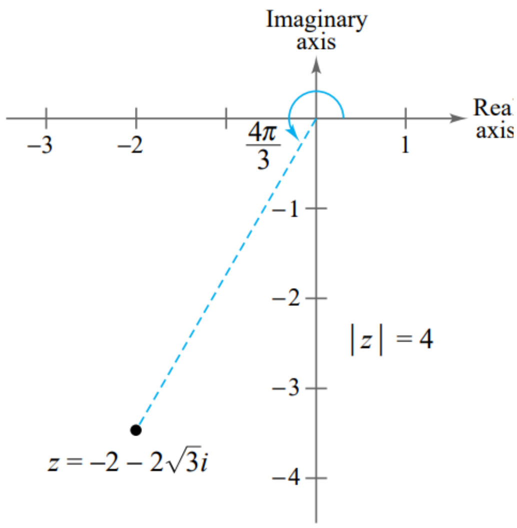
$$\tan\frac{\pi}{3} = \sqrt{3}$$

Because $Z = -2 - 2\sqrt{3}\,i$ lies in Quadrant III, then

$\theta = \pi + \frac{\pi}{3} = \frac{4}{3}\pi$

And the polar form of the given complex number will be

$$Z = r(cos\ \theta + i\ sin\ \theta) = 4(\cos\frac{4}{3}\pi + i\sin\frac{4}{3}\pi)$$



Multiplication and Division of Complex Numbers in the Polar Form

The polar form adapts nicely to multiplication and division of complex numbers. Suppose you are given two complex numbers

$$Z_1 = r_1(cos\ \theta_1 + i\ sin\ \theta_1)\ and\ Z_2 = r_2(cos\ \theta_2 + i\ sin\ \theta_2)$$

The product of Z_1 *and* Z_2 will be

$$Z_1Z_2 = r_1r_2(cos\ \theta_1 + i\ sin\ \theta_1)\ (cos\ \theta_2 + i\ sin\ \theta_2) =$$

$$Z_1Z_2 = r_1r_2[(cos\ \theta_1 cos\ \theta_2 - \ sin\ \theta_1 sin\ \theta_2) + i\ (sin\ \theta_1 cos\ \theta_2 + cos\ \theta_1\ sin\ \theta_2)]$$

Using the sum and difference formulas for cosine and sine, you can rewrite this equation as

$$Z_1Z_2 = r_1r_2[cos\ (\theta_1 + \ \theta_2) + i\ sin\ (\theta_1 + \ \theta_2)]$$

In the same manner the Quotient of Z_1 *and* Z_2 will be

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2}[cos\ (\theta_1 - \ \theta_2) + i\ sin\ (\theta_1 - \ \theta_2)], Z_2 \neq 0$$

Example

If $Z_1 = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$ $\quad and\ Z_2 = 8(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6})$

$Find\ (Z_1.Z_2)$

Sol.

$$Z_1.Z_2 = \left[2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)\right]\left[8\left(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}\right)\right]$$

$$Z_1Z_2 = r_1r_2[cos\ (\theta_1 + \ \theta_2) + i\ sin\ (\theta_1 + \ \theta_2)]$$

$$Z_1Z_2 = 16\left[\cos\left(\frac{2\pi}{3} + \frac{11\pi}{6}\right) + i\sin\left(\frac{2\pi}{3} + \frac{11\pi}{6}\right)\right]$$

$$Z_1Z_2 = 16\left[\cos\left(\frac{5\pi}{2}\right) + i\sin\left(\frac{5\pi}{2}\right)\right] = 16\left[\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right]$$

$$Z_1Z_2 = 16[0 + i] = 16i$$

Example

If $Z_1 = 24(\cos 300^\circ + i\sin 300^\circ)$ $\quad and\ Z_2 = 8(\cos 75^\circ + i\sin 75^\circ)$

$Find\ (\frac{Z_1}{Z_2})$

Sol.

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2}[cos\ (\theta_1 - \ \theta_2) + i\ sin\ (\theta_1 - \ \theta_2)] = \frac{24}{8}[\cos 225^\circ + i\sin 225^\circ]$$

$$\frac{Z_1}{Z_2} = 3\left[\frac{-\sqrt{2}}{2} + i\frac{-\sqrt{2}}{2}\right] = \frac{-3\sqrt{2}}{2} - i\frac{3\sqrt{2}}{2}$$

<u>Powers and Roots of Complex Numbers and DeMoivre's Theorem</u>

To raise a complex number to a power, consider repeated use of the multiplication rule.

$Z = r(cos\ \theta + i\ sin\ \theta)$

$Z^2 = r^2(cos\ 2\theta + i\ sin\ 2\theta)$

$Z^3 = r^3(cos\ 3\theta + i\ sin\ 3\theta)$

This pattern leads to the following important theorem, which is named after the French mathematician Abraham DeMoivre (1667–1754).

DeMoivre's Theorem

If $Z = r(cos\ \theta + i\ sin\ \theta)$ is a complex number and *n* is a positive integer, then $Z^n = r^n(cos\ n\theta + i\ sin\ n\theta)$

Example

Use DeMoivre's Theorem to find

$\left(-1+\sqrt{3}i\right)^{12}$

Sol.

$$\left(-1+\sqrt{3}i\right)^{12} = r^{12}(cos\ 12\theta + i\ sin\ 12\theta)$$

The absolute value of *Z* is

$$r = |a+bi| = \sqrt{a^2+b^2} = \sqrt{1+3} = 2$$

And the angle θ is given by

$$\tan\theta = \frac{b}{a} = \frac{-\sqrt{3}}{1} = -\sqrt{3}$$

$$Since\ \tan\frac{2\pi}{3} = -\sqrt{3}\quad then\ \theta = \frac{2\pi}{3}$$

$$-1+\sqrt{3}i = 2\left(cos\ \frac{2\pi}{3} + i\ sin\ \frac{2\pi}{3}\right)$$

$$\left(-1+\sqrt{3}i\right)^{12} = 2^{12}(cos\ 8\pi + i\ sin\ 8\pi) = 4096$$

Definition of *n*th Root of a Complex Number

The complex number $u = a + bi$ is an *n*th root of the complex number *z* if

$$z = u^n = (a+bi)^n$$

To find a formula for an *n*th root of a complex number, let *u* be an *n*th root of *z*, where

$$u = s(\cos\beta + i\sin\beta) and\ z = r(\cos\theta + i\sin\theta)$$

By DeMoivre's Theorem and the fact $u^n = z$

$$s^n(\cos n\beta + i \sin n\beta) = r(\cos\theta + i \sin\theta)$$

Taking the absolute values of both sides of this equation, it follows that $s^n = r$

Substituting back into the previous equation and dividing by *r*, you get

$$(\cos n\beta + i \sin n\beta) = (\cos\theta + i \sin\theta)$$

Thus, it follows that

$$\cos n\beta = \cos\theta \;\; and \;\; \sin n\beta = \sin\theta$$

Because both sine and cosine have a period of 2π these last two equations have solutions if and only if the angles differ by a multiple of 2π. Consequently, there must exist an integer *k* such that

$$n\beta = \theta + 2\pi k$$

Or

$$\beta = \frac{\theta + 2\pi k}{n}$$

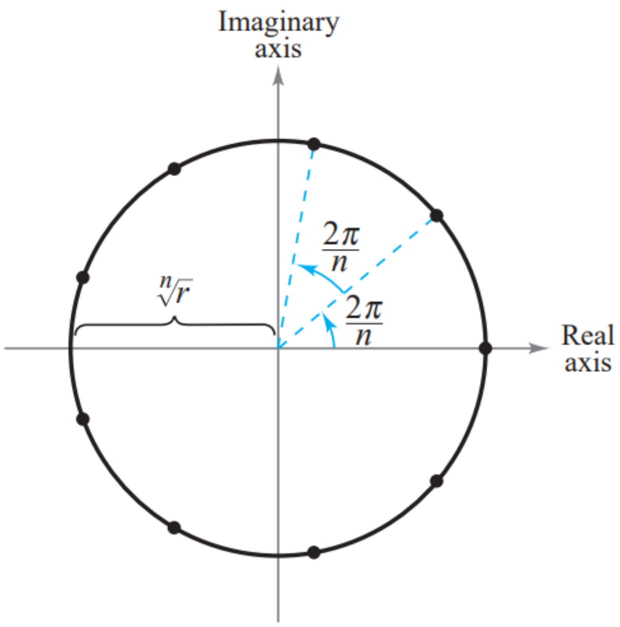
By substituting this value for *β* into the polar form of *u*, you get the following result:

THEOREM *n*th Roots of a Complex Number

For a positive integer *n*, the complex number $z = r(\cos\theta + i\sin\theta)$ has exactly *n* distinct *n*th roots given by

$$\sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right) \;\; where\ k = 0,1,2,\dots,n-1$$

This formula for the nth roots of a complex number *z* has a geometrical interpretation, as shown in the next Figure. Note that because the *n*th roots of *z* all have the same magnitude $\sqrt[n]{r}$ they all lie on a circle of radius $\sqrt[n]{r}$ with centre at the origin. Furthermore, because successive *n*th roots have arguments that differ by $\frac{2\pi}{n}$ the *n* roots are equally spaced along the circle.



Example

Find the three cube roots of

$$z = -2 + 2\,i$$

Solution Because *z* lies in Quadrant II, the polar form for *z*

$$z = -2 + 2\,i = \sqrt{8}\,(\cos 135^{\circ} + i \sin 135^{\circ})$$

By the formula for *n*th roots, the cube roots have the form

$$\sqrt[6]{8}\left(\cos\frac{135^{\circ} + 360^{\circ}k}{3} + i\sin\frac{135^{\circ} + 360^{\circ}k}{3}\right) \quad where\ k = 0,1,2$$

Then for $k = 0$

$$\sqrt{2}\,(\cos 45^{\circ} + i \sin 45^{\circ}) = 1 + i$$

For $k = 1$

$$\sqrt{2}\,(\cos 165^{\circ} + i \sin 165^{\circ}) \approx -1.366 + 0.366\,i$$

For $k = 2$

$$\sqrt{2}\,(\cos 285^{\circ} + i \sin 285^{\circ}) \approx 0.366 - 1.366\,i$$

Argand Diagram

Euler's Formula

The identity

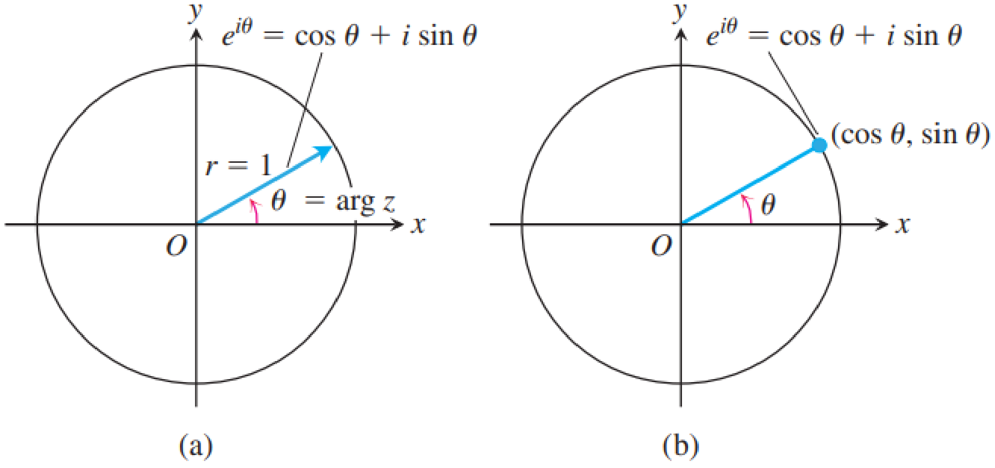
$$e^{i\theta} = \cos\theta + i\sin\theta$$

is called Euler's formula

Since $Z = r(\cos\theta + i\sin\theta)$, the Euler's formula enables us to rewrite this equation as

$$Z = re^{i\theta}$$

This formula, in turn, leads to the following rules for calculating products, quotients, powers, and roots of complex numbers. It also leads to Argand diagrams for $e^{i\theta}$. By taking $r = 1$, we can say that $e^{i\theta}$ is represented by a unit vector that makes an angle θ with the positive *x-axis*, as shown in Figure below:



Products

To multiply two complex numbers, we multiply their absolute values and add their angles.

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$

$$|z_1| = r_1 \ \ , \arg z_1 = \theta_1, |z_2| = r_2 \ \ , \arg z_2 = \theta_2$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$$

Example

If you know that

$$z_1 = \sqrt{2} e^{i\frac{\pi}{4}} \ , \ \ z_2 = 2e^{-i\frac{\pi}{6}}$$

Find $z_1 z_2$

Sol.

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)} = 2\sqrt{2} e^{i(\frac{\pi}{4}-\frac{\pi}{6})} = 2\sqrt{2} e^{i\frac{\pi}{12}}$$

$$z_1 z_2 = 2\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) \approx 2.73 + 0.73i$$

Quotients

Suppose $r_2 \neq 0$

Then $\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$

Example

If you know that

$$Z_1 = 1 + i \ , \ \ Z_2 = \sqrt{3} - i$$

Find z_1/z_2

Sol.

$$\left|\frac{Z_1}{Z_2}\right| = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$$

$$Z_1 = 1 + i = r_1 e^{i\theta_1}$$

$$a = 1, b = 1, r_1 = \sqrt{a^2 + b^2} = \sqrt{2}$$

$$\frac{b}{a} = \tan\theta_1 = 1, then\ \theta_1 = \frac{\pi}{4}$$

$$Z_1 = 1 + i = \sqrt{2} e^{i\frac{\pi}{4}}$$

$$Z_2 = \sqrt{3} - i = r_2 e^{i\theta_2}$$

$$a = \sqrt{3}, b = -1, r_2 = \sqrt{a^2 + b^2} = 2$$

$$\frac{b}{a} = \tan\theta_2 = \frac{-1}{\sqrt{3}} 1, then\ \theta_2 = \frac{-\pi}{6}$$

$$Z_2 = \sqrt{3} - i = 2e^{i\frac{-\pi}{6}}$$

$$\left|\frac{Z_1}{Z_2}\right| = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} = \frac{\sqrt{2}}{2} e^{i(\frac{\pi}{4} + \frac{\pi}{6})} = 0.707 e^{i\frac{5\pi}{12}}$$

$$0.707 e^{i\frac{5\pi}{12}} = 0.707(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12})$$

$$\left|\frac{Z_1}{Z_2}\right| \approx 0.183 + 0.683\, i$$

Powers

If (n) is a positive integer then

$$Z^n = Z.Z.Z \ldots Z \quad n\ factors$$

With

$$Z = re^{i\theta}$$

Then

$$Z^n = \left(re^{i\theta}\right)^n = r^n e^{in\theta}$$

$$r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta)$$

<u>Example</u>

If $Z = 4e^{i\frac{\pi}{3}}$

n=3

$$Z^3 = \left(re^{i\theta}\right)^3 = r^3 e^{3i\theta} = 4^3 e^{i\pi}$$